

Notes: Eeckhout & Sepahsalari (RES, 2024)

The Effect of Wealth on Worker Productivity

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1 Big picture

- **Question.** How do workers' asset holdings affect their job-search choices, job-finding probabilities, wages, and ultimately worker productivity?
- **Central mechanism.** Workers without much wealth use the labor market as self-insurance. They direct search toward jobs that are easier to get, even if those jobs are less productive and pay lower wages. Asset-rich workers can tolerate more unemployment risk and therefore search for higher-productivity, higher-wage jobs with lower job-finding probabilities.
- **Main concept.** The paper introduces a **precautionary job search motive**, parallel to the standard precautionary saving motive. Poor workers self-insure not only by trying to save but also by choosing safer submarkets in the labor market.
- **Theoretical contribution.** With consumption smoothing and decreasing absolute risk aversion, workers with higher assets sort into higher-productivity firms even when workers are equally skilled and there is no technological complementarity between worker skill and firm productivity.
- **Resolution of a directed-search puzzle.** The canonical directed-search model predicts that high wages come with longer queues, generating positive duration dependence. The paper shows that once unemployed workers run down assets, long-duration unemployed workers lower their search target; this generates **negative duration dependence** of reemployment wages.
- **Quantitative contribution.** The paper solves an infinite-horizon directed-search economy with risk-averse workers, endogenous savings, heterogeneous firms, and an ergodic asset distribution.
- **Calibration.** The model is calibrated to U.S. moments: unemployment, unemployment duration, replacement rate, vacancy cost, matching elasticity, asset dispersion, and dividend-output ratio.
- **Key quantitative result.** Workers with low assets search for lower-productivity jobs and find jobs faster. Workers with high assets search for higher-productivity jobs and have lower job-finding probabilities.
- **UI policy result.** Unemployment insurance improves consumption smoothing and encourages workers to search for more productive jobs, but it also lowers job-finding probabilities, reduces vacancy creation, lowers firm dividends, and raises unemployment.
- **Welfare result.** Aggregate welfare is inverted-U-shaped in UI benefits. Low benefits improve insurance more than they distort incentives; high benefits generate too much unemployment and lower output.
- **Productivity result.** Higher UI lowers total production but raises output per employed worker because workers search for and match with more productive firms.
- **Economic takeaway.** Wealth affects productivity because wealth affects risk tolerance in job search. Asset inequality therefore becomes productivity inequality, even among equally skilled workers.

2 Data, calibration, and model measurement

2.1 What the model measures

The model links three distributions:

- the distribution of workers' asset holdings;
- the distribution of firm productivities;
- the distribution of search targets, wages, and matching probabilities chosen by unemployed workers.

Interpretation: The key state variable is worker wealth. A worker with more assets can smooth consumption during a longer job search and therefore chooses a more productive but riskier submarket. A worker with few assets chooses a safer, lower-productivity submarket.

2.2 Workers and income risk

Workers are either unemployed or employed. Unemployed workers choose both savings and a search submarket. Employed workers choose savings and may separate exogenously.

- a_t is asset holdings.
- b is unemployment benefit.
- τ is a wage tax used to finance UI.
- d is firm dividends distributed to households.
- w_t is the wage paid by a filled job.
- R is the gross return on assets.
- λ is the exogenous separation probability.

Consumption if unemployed and employed is:

$$c_{u,t} = Ra_t - a_{t+1} + b + d, \quad (1)$$

$$c_{e,t} = Ra_t - a_{t+1} + (1 - \tau)w_t + d. \quad (2)$$

Interpretation: The same worker faces different consumption risk depending on assets. A worker close to the borrowing constraint cannot tolerate a long unemployment spell, so the job-search decision itself becomes an insurance tool.

2.3 Firms, productivity, and directed search

Firms differ in productivity y . A firm posts a wage w to attract applicants. The submarket is indexed by (y, w) and has tightness θ .

- Workers choose which submarket to search in.
- Higher-wage, higher-productivity jobs attract more competition and can have lower job-finding probabilities.
- Firms with higher productivity can offer higher wages while still making positive profits.

Interpretation: Directed search gives workers a menu of wage-risk bundles: safer low-wage jobs versus riskier high-wage jobs. Wealth determines which bundle is optimal.

2.4 Externally calibrated parameters

Table 1 externally calibrates the discount factor, interest rate, exogenous separation rate, borrowing constraint, benefit replacement rate, vacancy cost, and matching elasticity. The benchmark replacement rate is $b/w = 0.40$, and the unemployment rate is targeted at 6.1%.

Interpretation: These parameters discipline the quantitative environment. The replacement rate and vacancy cost are especially important for the UI exercises because they determine the trade-off between insurance and vacancy creation.

2.5 Endogenous outcomes matched by calibration

Table 2 reports model-versus-data moments such as unemployment, average benefit-wage ratio, unemployment duration, average matching probability, median assets relative to dividends, dividend-output ratio, matching elasticity, and vacancy cost share.

Interpretation: The model is not calibrated only to unemployment. It also targets asset and labor-market moments, which is crucial because the paper's mechanism is exactly the interaction between asset dynamics and job-search behavior.

3 Model and theoretical specifications

3.1 Unemployed worker problem

The unemployed worker chooses savings and a search submarket:

$$U(a_t) = \max_{a_{t+1}, y_{t+1}, w_{t+1}} \{u(c_{u,t}) + \beta [m(\theta_t)E(a_{t+1}, w_{t+1}) + (1 - m(\theta_t))U(a_{t+1})]\}, \quad (3)$$

subject to

$$c_{u,t} = Ra_t - a_{t+1} + b + d, \quad a_{t+1} \geq \underline{a}. \quad (4)$$

Interpretation: The unemployed worker chooses both financial self-insurance and labor-market self-insurance. Savings chooses resources for the future; the submarket choice chooses the risk and reward of job search.

3.2 Employed worker problem

The employed worker chooses savings:

$$E(a_t, w_t) = \max_{a_{t+1}} \{u(c_{e,t}) + \beta [\lambda U(a_{t+1}) + (1 - \lambda)E(a_{t+1}, w_{t+1})]\}, \quad (5)$$

subject to

$$c_{e,t} = Ra_t - a_{t+1} + (1 - \tau)w_t + d, \quad a_{t+1} \geq \underline{a}. \quad (6)$$

Interpretation: Employed workers save because they anticipate future job loss. The employed worker's asset accumulation feeds back into future unemployment search choices.

3.3 Firm value

A vacant firm of productivity y pays vacancy cost k and posts a wage:

$$V(y) = -k + \max_w \beta [q(\theta)J(y, w) + (1 - q(\theta))V(y)]. \quad (7)$$

A filled job has value

$$J(y, w) = y - w + \beta [\lambda V(y) + (1 - \lambda)J(y, w)]. \quad (8)$$

Interpretation: Firms face the other side of the directed-search trade-off. Higher wages attract workers and raise filling probabilities, but reduce profits from a filled match.

3.4 Two-period model and sorting

In the two-period economy, all workers begin unemployed. The unemployed worker solves:

$$U(a_1) = \max_{a_2, y, w} \{u(Ra_1 - a_2) + \beta [m(\theta)u(Ra_2 + w) + (1 - m(\theta))u(Ra_2)]\}. \quad (9)$$

Proposition 1 shows that higher initial assets lead workers to apply to higher-productivity jobs under a condition implied by decreasing absolute risk aversion (DARA).

Interpretation: This result is the cleanest statement of the paper’s mechanism. Richer workers can tolerate the risk of long queues; poorer workers choose safer jobs. The sorting is preference-based, not technology-based.

3.5 Infinite-horizon sorting

In the infinite-horizon economy, the sorting condition uses value functions rather than only primitives:

$$\frac{E_{a_{t+1}}(a_{t+1}, y) - U_{a_{t+1}}(a_{t+1})}{E(a_{t+1}, y) - U(a_{t+1})} < \frac{E_{w_{t+1}a_{t+1}}(a_{t+1}, y)}{E_{w_{t+1}}(a_{t+1}, y)}. \quad (10)$$

Interpretation: In the infinite horizon, assets affect today’s search choice and tomorrow’s asset distribution. The equilibrium must be solved jointly with the stationary distribution of unemployed and employed workers.

3.6 UI policy counterfactuals

The government pays UI benefit b to unemployed workers and finances it with a wage tax τ . The policy experiments vary b and solve the new steady state.

Interpretation: UI changes the worker’s outside option, search target, firm vacancy creation, wages, profits, taxes, dividends, and asset accumulation. The welfare effect is general equilibrium, not merely a transfer to unemployed workers.

4 Identification and quantitative design

4.1 What identifies the mechanism

- The model creates a mapping from wealth to productivity choice, job-finding probability, and wages.
- The mechanism is validated against empirical patterns: asset-poor workers have higher job-finding rates and lower reemployment wages.
- The model also matches the negative duration dependence of reemployment wages.

Interpretation: The empirical content is not a randomized experiment. It is a quantitative validation that the model reproduces robust labor-market patterns that the canonical directed-search model struggles to explain.

4.2 Why the mechanism is economically meaningful

- Low assets make unemployment costly in consumption terms.
- The worker responds by choosing a submarket with higher job-finding probability.
- That submarket has lower productivity and lower wages.
- Therefore, liquidity constraints distort the allocation of workers across firms.

Interpretation: The paper connects household finance to labor productivity. Wealth is not just a consumption-smoothing buffer; it changes the quality of the job a worker chooses.

4.3 What the UI exercise isolates

- UI improves consumption smoothing.
- UI changes job-search risk taking by increasing the outside option.
- UI raises wages and reduces firm profits.
- UI reduces vacancies and job-finding rates.
- UI raises average productivity of employed workers but lowers total output.

Interpretation: The welfare optimum balances insurance and productivity-search benefits against lower vacancy creation, lower dividends, and higher unemployment.

4.4 Limitations

- The model abstracts from many observable worker types such as education, occupation, health, and family structure.
- The main heterogeneity is asset holdings, so other channels of duration dependence are not modeled.
- UI is modeled as a constant benefit financed by a wage tax, not as a fully optimal mechanism with private information.
- The welfare comparisons depend on the calibrated utility function, matching technology, and dividend distribution assumptions.

Interpretation: These limitations make the paper best understood as a mechanism and quantitative benchmark, not as a complete optimal unemployment-insurance design exercise.

5 Tables and figures

5.1 Tables 1–2: Calibration and endogenous outcomes

Read:

- Table 1 reports externally calibrated parameters: discount factor 0.9976, monthly interest rate 0.002, separation rate 0.017, borrowing constraint -1 , replacement rate 0.85, vacancy cost 0.52, and matching elasticity 0.41.
- Table 2 shows that the model matches unemployment at 6.1%, average replacement rate 0.40, unemployment duration around 4 months versus 4.50 in the data, and a monthly matching probability of 0.25%.
- The model also matches the vacancy-cost share and dividend-output ratio closely.

Interpretation: The calibration deliberately anchors both household-side and labor-market-side moments. This is important because the mechanism requires a credible interaction between wealth accumulation and search risk. If the asset distribution were poorly matched, the wealth-productivity channel would be unconvincing; if unemployment duration or matching rates were poorly matched, the directed-search channel would be unconvincing.

Economic message: Tables 1–2 establish the quantitative baseline against which the UI experiments are evaluated.

What not to over-read: The table does not prove the mechanism. It shows that the benchmark economy can reproduce key moments while embedding the proposed sorting channel.

TABLE 1 Externally calibrated parameters			
	Definition	Value	Source
β	Discount factor	0.9976	Krusell <i>et al.</i> (2019)
r	Interest rate	0.002	Yearly risk-free rate of 2.5%
λ	Exogenous separation	0.017	Eeckhout and Lindenlaub (2019) (CPS)
$\underline{a}$	Borrowing constraint	-1	
b	Replacement rate	0.85	Average $\frac{b}{w} = 0.4$, Shimer (2005)
k	Cost of vacancy	0.52	$\frac{k}{y} = 21\%$, Shimer (2005)
γ	Elasticity of matching function	0.41	Job finding probability—CPS

TABLE 2 Endogenous outcomes			
Moments	Model	Data	Source
$u\%$	6.1%	6.1%	Federal Reserve Bank of St. Louis (FRED)
avg. $(\frac{b}{w})$	0.40	0.40	Shimer (2005)
unemp. dur.	4	4.50	Federal Reserve Bank of St. Louis (FRED)
avg. $m(\theta)$	0.25%	0.25%	CPS from Eeckhout and Lindenlaub (2019)
med $(\frac{a}{w})$	0.49	0.46	PSID
$\frac{\text{tot. dividend}}{\text{tot. output}}$	5.3%	4.8%	Federal Reserve Bank of St. Louis (FRED)
elasticity of $m(\theta)$ to θ	0.40	[0.27–0.50]	Pissarides (2009) & Shimer (2005)
avg. $(\frac{k}{y})$	0.21%	0.21%	Shimer (2005)

Tables 1–2: calibrated parameters and endogenous outcomes.

5.2 Figure 1: Equilibrium allocation and wages

Read:

- Panel (a) shows that workers with higher assets search for higher-productivity jobs.
- Panel (b) shows that wages at hiring rise with worker assets.

- The paper reports that high-asset workers choose jobs with productivity around 2.50 versus around 2.39 for low-asset workers.

Interpretation: This is the first direct visualization of the wealth-productivity mechanism. Asset-rich workers are not more skilled in the model; they are more financially able to tolerate a lower job-finding probability. Their wealth changes their chosen search target. Thus, differences in worker productivity arise from portfolio/wealth differences, not from intrinsic productivity differences.

Economic message: Wealth inequality creates job-productivity inequality. Liquidity-poor workers sort into low-productivity jobs because they need employment insurance.

What not to over-read: The wage differences in the figure look small in levels, but the paper stresses that they are paid over long employment spells. Small wage differences can therefore imply large lifetime income differences.

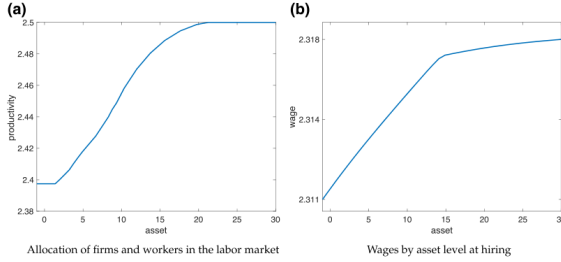


FIGURE 1
Equilibrium allocation and wages



FIGURE 2
Policy functions of the unemployed

5.3 Figure 2: Policy functions of the unemployed

Read:

- Panel (a) shows that low-asset unemployed workers have higher job-finding probabilities than high-asset unemployed workers.
- Panel (b) compares savings choices for employed and unemployed workers relative to the 45-degree line.
- Panel (c) shows asset changes: unemployed workers run down assets; employed workers accumulate assets.

Interpretation: Figure 2 links the two insurance margins. Poor unemployed workers use labor-market insurance by choosing high-probability jobs. At the same time, unemployment depletes assets, forcing workers to search for safer and lower-productivity jobs if unemployment persists. Employed workers rebuild financial self-insurance because they anticipate future job loss.

Economic message: The model contains a feedback loop: unemployment reduces assets; lower assets reduce the target productivity of job search; lower target productivity raises job-finding probability but lowers future wages.

What not to over-read: The job-finding probability is endogenous. It should not be interpreted as an exogenous worker type or fixed employability characteristic.

5.4 Figure 3: Ergodic distributions

Read:

- Panel (a) shows the stationary asset distribution among unemployed workers.
- The distribution is concentrated at low and middle assets, with little mass at very high assets.
- Panel (b) shows the distribution of vacant firms across productivity levels.
- The vacant-firm distribution is tilted toward lower productivity firms, reflecting endogenous firm entry and filling probabilities.

Interpretation: Figure 3 shows that the individual policy functions in Figure 2 aggregate into a stationary economy. Unemployed workers repeatedly run down assets, while employed workers rebuild them. On the firm side, high-productivity firms are more attractive and have different filling incentives, affecting which vacancies remain open.

Economic message: The steady state is not simply a fixed distribution of workers plus exogenous jobs. Both the worker wealth distribution and the composition of vacancies are equilibrium objects.

What not to over-read: These are densities in the calibrated model, not direct empirical histograms. Their role is to show the internal consistency of the model economy.

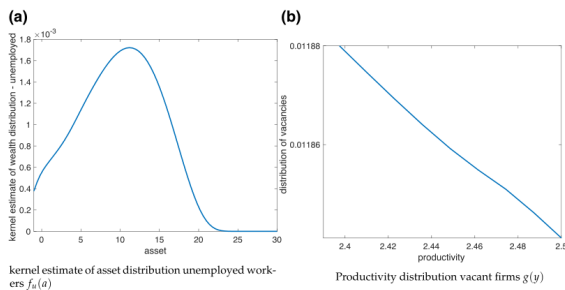


FIGURE 3
Ergodic distributions (densities)

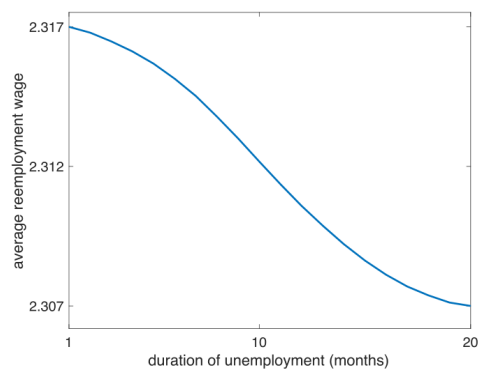
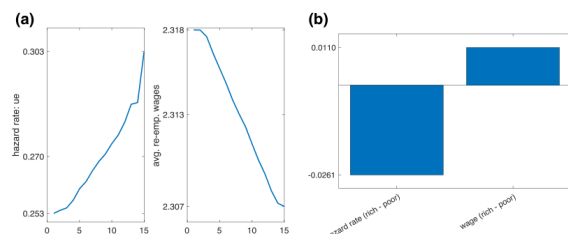


FIGURE 4
Unemployment duration & reemployment wages



5.5 Figures 4–5: Duration dependence

Read:

- Figure 4 shows that average reemployment wages fall with unemployment duration.
- Figure 5(a) decomposes time-series duration effects: as unemployment duration rises, the hazard rate rises while reemployment wages fall.

- Figure 5(b) shows cross-sectional differences between rich and poor workers: rich workers have lower job-finding rates but higher wages.
- The paper reports a strong negative correlation between hazard rates and reemployment wages over time.

Interpretation: These figures are central because they solve a long-standing problem for directed-search models. Canonical directed search predicts higher wages come with longer queues, so it tends to produce positive duration dependence. Here, unemployed workers lose assets over time, so their optimal search target moves downward. Over time, this wealth-depletion effect dominates the canonical queueing effect, producing negative duration dependence in wages.

Economic message: A declining reemployment wage is not imposed. It arises because unemployed workers become poorer and therefore choose safer, lower-wage jobs as the spell continues.

What not to over-read: The model does not deny that skill depreciation, employer screening, or stigma may also create negative duration dependence. It shows that asset depletion alone can generate a similar pattern.

5.6 Figure 6: Benefits, consumption, search allocation, and job finding

Read:

- Panel (a) shows that unemployed consumption rises with UI benefits, especially for low-asset workers.
- Panel (b) shows that higher benefits shift workers toward more productive job targets.
- Panel (c) shows that higher benefits lower job-finding probabilities.

Interpretation: UI changes both consumption and search behavior. For asset-poor workers, the direct insurance effect is large: benefits relax the need to accept low-productivity jobs quickly. With more insurance, workers search for better jobs. But this increases competition for attractive jobs and lowers matching probabilities.

Economic message: UI has an intensive productivity effect: it pushes workers toward higher-productivity jobs. But this comes at the cost of slower matching.

What not to over-read: The increase in productivity targeting is not equivalent to higher total production. Fewer matches and fewer vacancies can offset or dominate the productivity gain.

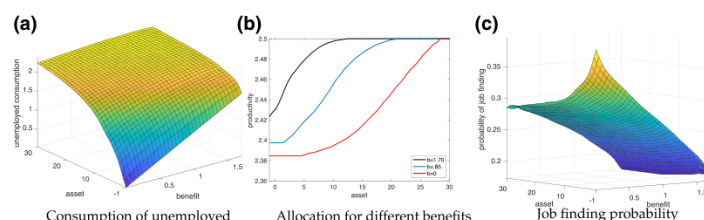


FIGURE 6
Consumption, equilibrium allocation, and job finding probability for different levels of benefits

(2010) with random search, though now with two-sided heterogeneity, a change in unemployed

Figure 6: consumption, allocation, and job-finding probability across benefit levels.

5.7 Figures 7–8: UI, unemployment, vacancies, dividends, and value of unemployment

Read:

- Figure 7(a) shows unemployment increasing with UI benefits.
- Figure 7(b) shows total vacancies declining as benefits rise.
- Figure 7(c) shows firm dividends falling with benefits.
- Figure 8(a) shows the value of unemployment over assets and benefits.
- Figure 8(b) shows that the value gap between asset-rich and asset-poor unemployed workers changes with benefits.
- Figure 8(c) shows optimal benefits by asset level and employment status.

Interpretation: Figure 7 makes the general equilibrium cost visible. Benefits improve insurance, but they raise workers' outside options, reduce firms' surplus, and lower vacancy creation. Figure 8 shows heterogeneity in preferences over UI: low-asset unemployed workers value benefits most, while employed and asset-rich workers face larger costs through taxes, lower dividends, and lower matching efficiency.

Economic message: UI creates a distributional conflict within both unemployed and employed groups. The asset-poor want more insurance; the asset-rich bear more of the equilibrium cost.

What not to over-read: The policy trade-off is not simply unemployed versus employed. Asset holdings matter within each employment state.

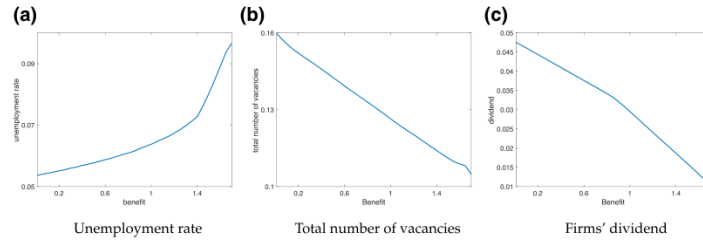


FIGURE 7
Equilibrium unemployment, vacancy creation and dividends for different levels of benefits

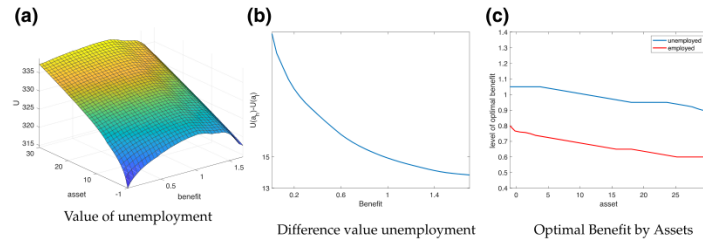


FIGURE 8
Value of unemployment

is much more pronounced for the low asset holders. Benefits induce them to compete for high

Figures 7–8: UI effects on unemployment, vacancies, dividends, and value of unemployment.

5.8 Figures 9–10 and Table 3: Welfare

Read:

- Figure 9 shows welfare measures for unemployed and employed workers at low, medium, and high assets.
- Figure 10 aggregates welfare across the benchmark distribution and shows an inverted-U shape in benefits.
- Table 3 reports welfare changes relative to the benchmark benefit level $b = 0.85$.
- Moving from zero benefits to the benchmark raises welfare for many workers, especially low-asset unemployed workers.
- Moving to very high benefits lowers welfare, especially for employed and asset-rich workers.

Interpretation: The inverted-U is the quantitative summary of the paper. At low benefit levels, additional UI mostly improves consumption smoothing and allows better search. At high benefit levels, lower vacancies, lower dividends, higher unemployment, and lower output dominate. Table 3 shows that aggregate averages hide sharp heterogeneity by assets and employment status.

Economic message: The optimal UI level is not uniform across workers. Poor unemployed workers benefit more from increases in UI, while rich employed workers face larger losses when benefits become too generous.

What not to over-read: The welfare maximum is not a universal policy prescription. It is conditional on the calibrated model, the assumed constant benefit, the tax financing rule, and the stationary distribution.

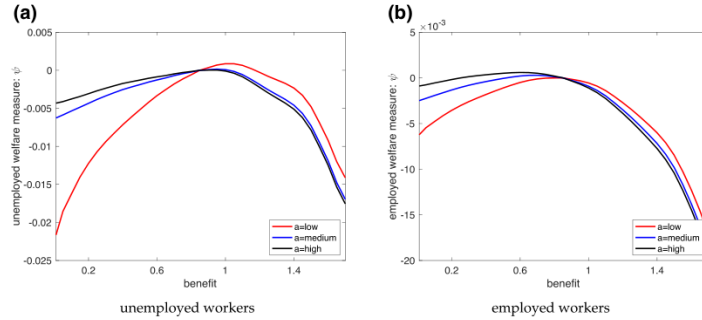
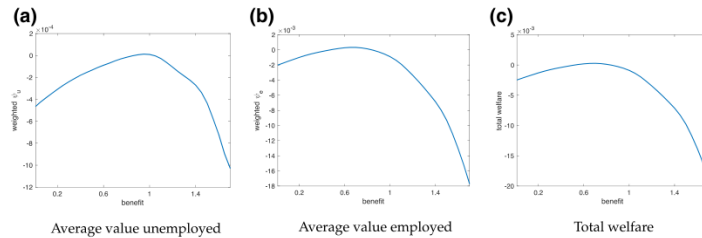


FIGURE 9
Welfare measure: ψ



Figures 9–10: individual and aggregate welfare as UI benefits vary.

TABLE 3
Welfare change compared to benchmark economy (first three columns are welfare gains for all workers, the unemployed, and the employed. The last six columns illustrate the welfare gains by an unemployed and employed worker with low(l), medium(m) and high(h) asset holding)

from $b = 0.8$ to $b =$	Total %	Unemp. %	Emp. %	$a_{u,l}$ %	$a_{u,m}$ %	$a_{u,h}$ %	$a_{e,l}$ %	$a_{e,m}$ %	$a_{e,h}$ %
0	-0.25	-0.05	-0.20	-2.16	-0.63	-0.43	-0.62	-0.25	-0.00
0.10	-0.19	-0.04	-0.15	-1.58	-0.54	-0.39	-0.47	-0.19	-0.00
0.35	-0.06	-0.02	-0.04	-0.81	-0.29	-0.20	-0.22	-0.06	0.00
0.45	-0.02	-0.02	-0.01	-0.62	-0.22	-0.14	-0.15	-0.02	0.00
0.70	0.03	-0.01	0.04	-0.18	-0.07	-0.05	-0.01	0.04	0.00
0.85	0	0	0	0	0	0	0	0	0
0.95	-0.05	0.00	-0.05	0.08	0.02	0.01	-0.02	-0.05	-0.00
1.10	-0.18	-0.00	-0.18	0.10	-0.04	-0.07	-0.12	-0.18	-0.00
1.30	-0.52	-0.02	-0.50	-0.13	-0.32	-0.37	-0.42	-0.52	-0.00
1.55	-1.10	-0.05	-1.05	-0.60	-0.86	-0.92	-0.96	-1.11	-1.00
1.70	-1.87	-0.10	-1.77	-1.41	-1.70	-1.75	-1.71	-1.87	-1.50

Table 3: welfare changes relative to the benchmark UI economy.

5.9 Figure 11: Total output and productivity

Read:

- Panel (a) shows that total output falls as benefits rise.
- Panel (b) shows that output per worker rises as benefits rise.
- The paper emphasizes that higher benefits shift workers toward more productive jobs but reduce the total number of filled jobs.

Interpretation: This figure distinguishes productivity per match from aggregate production. Better-insured workers search for better jobs, so average productivity of employed workers rises. But job-finding probabilities and vacancy creation fall, so fewer workers are employed and total output declines.

Economic message: UI can raise productivity per worker while reducing total output. The productivity gain is an intensive-margin effect; the output loss is an extensive-margin effect.

What not to over-read: Higher productivity per worker should not be read as an unambiguous welfare gain. Welfare depends on consumption smoothing, joblessness, taxes, dividends, and matching probabilities.

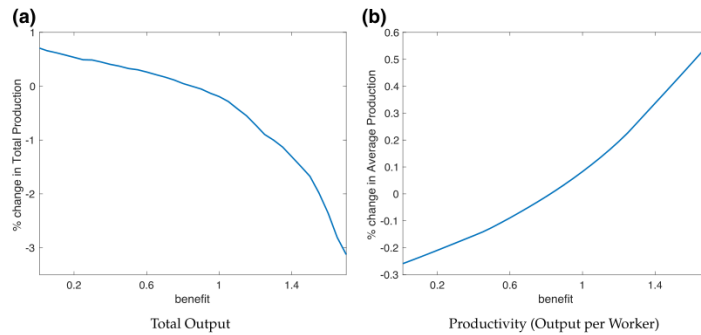


FIGURE 11
Total output and productivity

Figure 11: total output and productivity.

5.10 Appendix Tables B.1–B.3: Empirical and model regressions

Read:

- Table B.1 reports duration-dependence regressions using NLSY data and controls.
- Table B.2 reports the model regression of job-finding rates on transformed assets.
- Table B.3 reports the model regression of reemployment wages on unemployment duration.
- The model produces a negative relation between assets and job-finding rates and a negative relation between duration and reemployment wages.

Interpretation: The appendix regressions connect the model’s internal objects to empirical moments. They are not the main calibration tables, but they validate that the model can reproduce two important signs: wealthier workers find jobs more slowly, and longer unemployment spells lead to lower reemployment wages.

Economic message: The model is not only qualitatively plausible; it produces regression patterns consistent with observed labor-market evidence.

What not to over-read: These regressions are reduced-form validations. They do not identify all possible channels of duration dependence or wealth effects.

TABLE B.1
Duration dependence regression

	All	
	(1)	(2)
Unemp. Duration	0.0013 (0.0004)	0.0012 (0.0004)
Liq. Wealth		0.0089 (0.0016)
Observations	14381	14381

TABLE B.2
Regression of the job finding rate on assets

ω_0	0.27 (0.00004)
ω_1	-0.71 (0.00001)
R^2	0.82
N	49314

Tables B.1–B.2: NLSY/data and model regressions.

TABLE B.3
Regression of wages on duration

ζ_0	2.31 (0.000007)
ζ_1	-0.0006 (0.000001)
R^2	0.34
N	307257

Table B.3: wages on duration regression.

5.11 Appendix Figures D.1–D.4: Dividend-distribution robustness

Read:

- Figures D.1–D.2 repeat welfare calculations under an alternative dividend distribution assumption.
- Figures D.3–D.4 show that welfare remains inverted-U-shaped, but optimal benefits can move higher.
- The alternative dividend regime changes who bears the dividend cost of higher UI.

Interpretation: The dividend distribution matters because UI changes firm profits. If dividends are distributed differently across workers, the incidence of the UI-induced profit decline changes. The core mechanism remains, but the welfare peak shifts because the cost side of UI is redistributed.

Economic message: Robustness exercises confirm that the insurance/search trade-off remains even when profit incidence changes.

What not to over-read: The level of optimal UI is sensitive to incidence assumptions. The robust conclusion is the mechanism and inverted-U shape, not one exact benefit level.

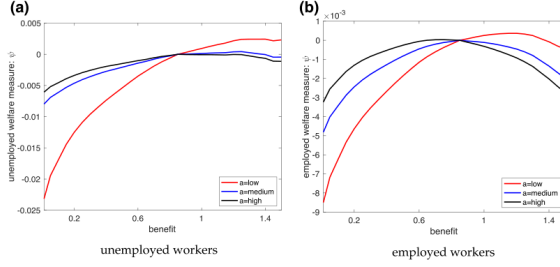


FIGURE D.1
Welfare measure: ψ

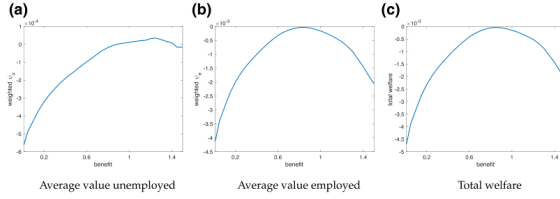


FIGURE D.2
Welfare

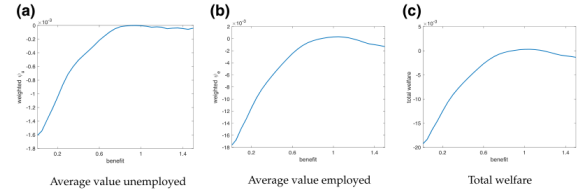
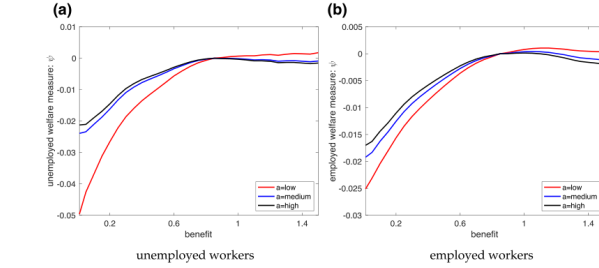


FIGURE D.4
Welfare



5.12 Appendix Figures E.1–E.2: Endogenous capital and interest rate

Read:

- Figures E.1–E.2 repeat the welfare analysis when capital and the interest rate are endogenous.
- The qualitative welfare shapes remain similar to the benchmark.
- Welfare maxima occur at higher benefit levels because the endogenous interest-rate response changes saving incentives.

Interpretation: Endogenous capital modifies the savings channel. Lower interest rates reduce the return to self-insurance and strengthen the relative value of external insurance. This pushes the welfare optimum toward higher UI but does not overturn the main mechanism.

Economic message: The UI trade-off is robust to capital-market general equilibrium effects.

What not to over-read: The endogenous-capital extension is not the main calibrated economy. It is a robustness check showing that the results do not rely entirely on the small-open-economy interest-rate assumption.

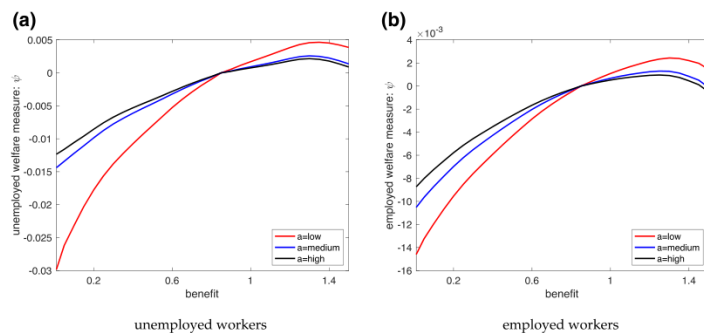
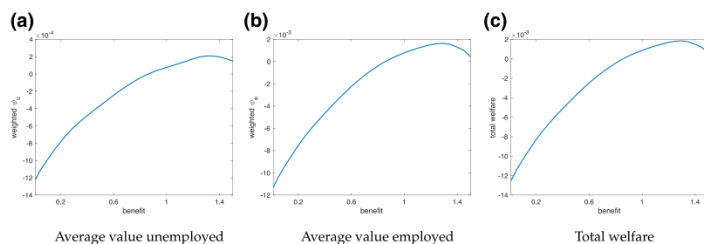


FIGURE E.1
Welfare measure: ψ when capital is endogenous



Figures E.1–E.2: welfare with endogenous capital and interest rate.

5.13 Appendix Figure F.1: Shift in productivity distribution

Read:

- Panel (a) shows that a shift in the productivity distribution moves allocations upward or downward.
- Panel (b) shows that changes in job-finding probabilities are strongest for low-asset workers.
- Panel (c) shows that changes in wages are positively related to assets.

Interpretation: The figure shows how asset holdings mediate aggregate productivity shocks. When productivity improves, asset-rich workers capture more wage gains because they are willing to search for better jobs. Low-asset workers benefit more through job-finding probabilities because they still value search safety.

Economic message: Aggregate productivity changes have distributional effects through the same wealth-search channel that drives the baseline model.

What not to over-read: The figure is a comparative static, not a full business-cycle model. It shows how the allocation changes when the productivity distribution shifts.

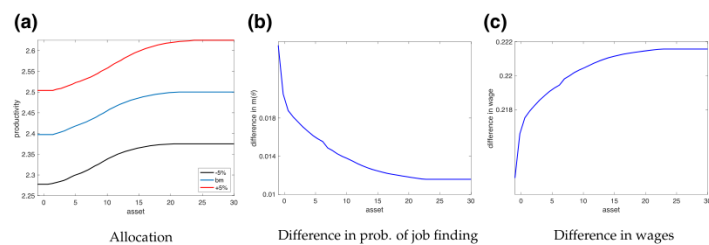


FIGURE F.1
Change in productivity distribution

across the distribution of assets. The change in probability of job finding is least for high asset

Figure F.1: changing the productivity distribution.

6 Short summary

- The paper studies how wealth affects worker productivity in a frictional labor market.
- Workers use both savings and job-search direction to self-insure against unemployment risk.
- Asset-poor workers search for safer, lower-productivity jobs with higher job-finding probabilities.
- Asset-rich workers search for riskier, higher-productivity jobs with higher wages.
- This creates positive sorting between worker wealth and firm productivity.
- Consumption smoothing resolves a weakness of canonical directed-search models by generating negative duration dependence of reemployment wages.
- The calibrated economy matches key unemployment, duration, matching, asset, and dividend moments.
- UI improves insurance and encourages workers to search for more productive jobs.
- Higher UI also lowers job-finding probabilities, vacancies, firm dividends, and total output.
- Welfare is inverted-U-shaped in benefits.
- Low-asset unemployed workers gain the most from higher UI.
- Asset-rich employed workers bear more of the cost when benefits become too generous.
- Higher benefits raise output per employed worker but reduce total production.
- The paper links household wealth inequality to labor productivity inequality.

7 Conclusion

7.1 Core conclusion

Eeckhout and Sepahsafari show that asset holdings affect worker productivity because workers use job search as self-insurance. When workers lack assets, unemployment is costly in consumption terms. These workers therefore search for jobs that are easier to obtain, even though those jobs have lower productivity and pay lower wages. Asset-rich workers can afford longer unemployment spells and search for higher-productivity jobs.

7.2 Economic intuition

The central intuition is that wealth changes risk tolerance in the labor market. A low-wealth unemployed worker faces a binding or nearly binding consumption-smoothing problem. The worker therefore values a high job-finding probability. A high-wealth worker values the wage and productivity upside more because wealth buffers the risk of remaining unemployed. This converts financial wealth into labor-market productivity.

7.3 The paper's main theoretical contribution

The paper shows that positive sorting between wealth and firm productivity arises without skill heterogeneity or technological complementarity. The complementarity is preference-based: high assets make risky, high-productivity jobs more attractive. This is important because it means that productivity differences can arise from financial conditions rather than human capital alone.

7.4 Contribution to directed-search theory

Canonical directed search has trouble generating negative duration dependence in wages because high wages attract more applicants and therefore imply longer queues. This paper reverses the overall implication through consumption smoothing. As unemployment duration rises, workers run down assets and lower their search target. The asset-depletion channel dominates the canonical queueing channel in the calibrated economy.

7.5 Policy conclusion on unemployment insurance

UI has both insurance and incentive effects. At low benefit levels, increasing UI raises welfare because it improves consumption smoothing and allows unemployed workers to search for better jobs. At high benefit levels, the costs dominate: job-finding probabilities fall, vacancy creation falls, dividends fall, unemployment rises, and total output declines. This generates an inverted-U welfare schedule.

7.6 Distributional conclusion

The welfare effects of UI are highly heterogeneous. Low-asset unemployed workers benefit most from higher UI. Asset-rich employed workers are more exposed to the costs through taxes, lower dividends, and worse equilibrium job creation. The paper therefore shows a conflict not only between employed and unemployed workers, but also between asset-poor and asset-rich workers within each employment state.

7.7 Productivity conclusion

Higher UI can raise output per employed worker because it improves workers' ability to search for more productive jobs. However, this productivity improvement occurs alongside fewer total matches and lower total output. The distinction between productivity per worker and aggregate production is one of the paper's key policy insights.

7.8 Limitations and caveats

The model abstracts from many real-world dimensions of heterogeneity, such as worker skills, health, occupations, family constraints, and search networks. It also studies a simple UI system rather than a fully optimal mechanism. These simplifications are useful for isolating the asset-search channel, but they mean that the quantitative optimal benefit level should not be interpreted as a literal policy formula.

7.9 Takeaway

The lasting message is that wealth matters for productivity because wealth affects the jobs workers can afford to search for. Financial fragility can push equally skilled workers into lower-productivity jobs. UI partially offsets that distortion, but too much UI reduces job creation. The policy problem is therefore to provide enough insurance to improve search quality without undermining vacancy creation and aggregate output.